

Erdős Problem #456

EP-456

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let p_n be the smallest prime $\equiv 1 \pmod{n}$ and let m_n be the smallest integer such that $n \mid \phi(m_n)$.

Is it true that $m_n < p_n$ for almost all n ? Does $p_n/m_n \rightarrow \infty$ for almost all n ? Are there infinitely many primes p such that $p-1$ is the only n for which $m_n = p$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>